Istituto di Ricerca e Certificazione per le Costruzioni Sostenibili
Notified Body No. 1994 in accordance with Regulation CPR (EU) No. 305/2011

TEST REPORT

Number:

1994-CPR-RP2790

Issuing date:

23rd January 2024

Applicant:

Everest SHPK
Bulevardi Blu, n. 544
1030, Tirana, Albania

Tested product:

System nodes for aluminum casement windows, of the series of trade name "71 000"
(see description)

Executed tests:

Calculation of thermal transmittance **EN ISO 10077-2:2017**

Normative references:

Harmonized product standard **EN 14351-1:2006+A2:2016**

The results in the test report relate only to the tested samples identified in this test report.
The test results relate to the behavior of the tested samples under specific test conditions. They are not intended to be the only criteria for assessing safety in use and environmental performance. Under real conditions, the behavior of the product may differ from the performance analyzed under the standard test conditions. The test report is not intended as a certificate of product conformity and the laboratory is not responsible for the conformity of production with the sample tested.

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1 Description of the tested sample

The tested series consists of n°1 system node for aluminum casement windows, belonging to the series commercially named “71 000”.

Identification code assigned by the laboratory: **0910/23/IR**.

The description and technical drawings below, referring to the analyzed sample, have been provided by the applicant under his own responsibility. These data influence the result and as they cannot be technically verified, IRCCOS declines any responsibility for the validity of the results.

PRODUCT DESCRIPTION PROVIDED BY THE CUSTOMER		
• Typology:	Nodes for casement windows	
• Material:	Aluminum EN AW 6060 T6 (EN 573-3, EN 12020.2)	
• Profiles:	- frame - sash - glazing bead produced by Everest SHPK, Tirana, Albania.	item 71101, item 71281, item 10407,
• Glazing:	triple glazing 48 mm thickness, produced by Everest SHPK, Tirana, Albania.	
• Thermal break:	noryl profiles Alfatherm produced by Everest SHPK, Tirana, Albania.	item 3351, item 3361, item 3351
• Glazing gaskets:	- external glazing gasket in EPDM+EPDM sponge - internal glazing gasket in EPDM produced by Everest SHPK, Tirana, Albania.	item L407EP, item1020,
• Sealing gaskets:	- gasket in EPDM+EPDM sponge - gasket in EPDM - central gasket in EPDM full expanded produced by Everest SHPK, Tirana, Albania.	item L407EP, item L443EP, item L471EP,
• Insulating:	- under glass insulation in PE Polyethylene - insulation in PE Polyethylene - aluminum foil low emissivity produced by Everest SHPK, Tirana, Albania.	item EV005PE, item EV006PE, item EV001AF,
• Declared nominal dimensions:	see attached technical drawings.	

PRODUCT TECHNICAL DRAWINGS PROVIDED BY THE CUSTOMER

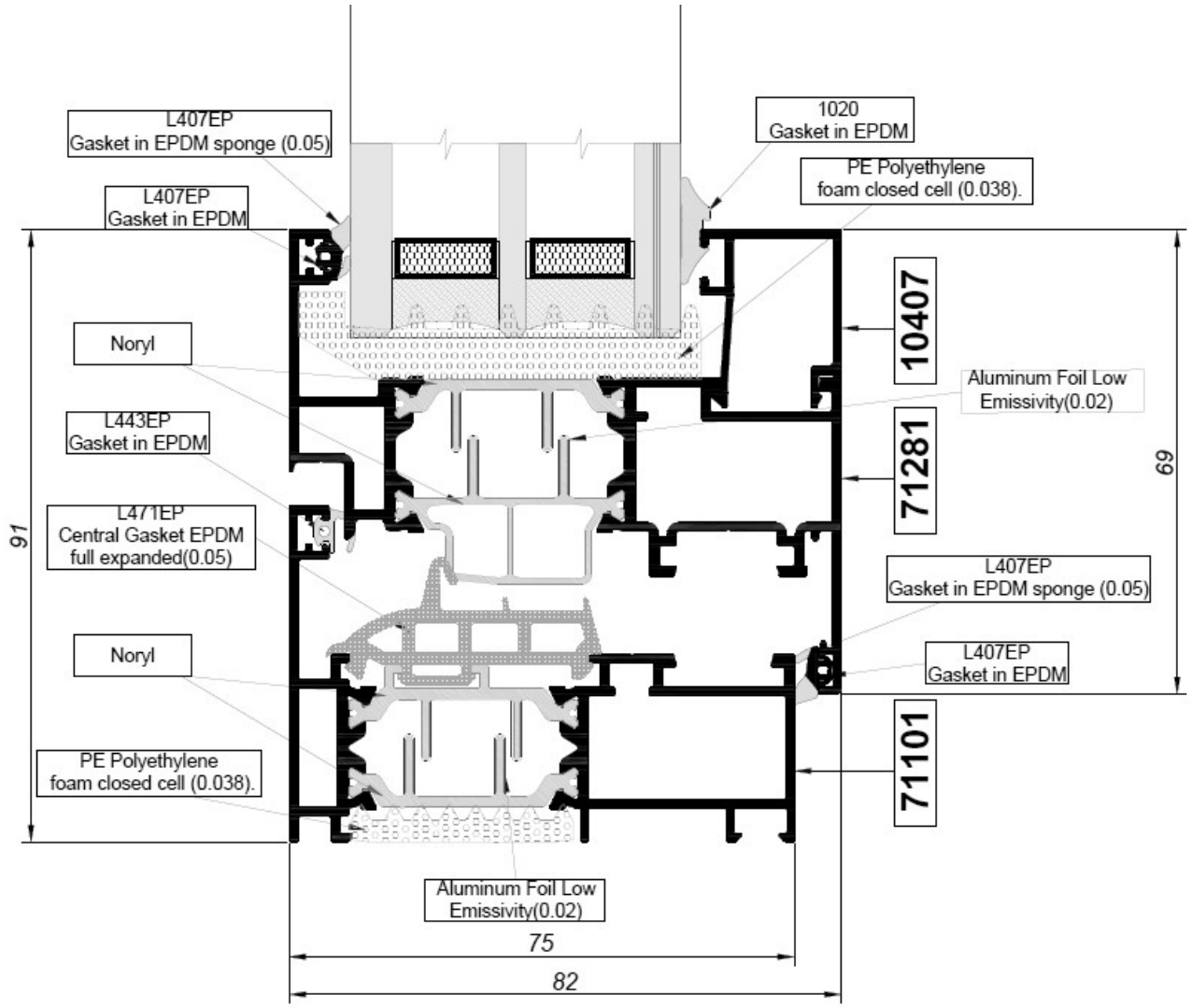


Fig. 1. Section of node 1 – Lateral / Lower / Upper node
(declared nominal dimensions in mm)

2 Methodology of analysis

2.1 Determination of thermal transmittance of frame nodes

The thermal transmittance calculation was performed in accordance with the standard EN ISO 10077-2:2017, using "Radiosity Method" (see § 6.4.2. of EN ISO 10077-2:2017), applying the procedure IOP 01 rev.01.

"Flixo 8.2" software was used for calculations. Materials were selected from the Flixo 8.2 software library based on documentation provided by the customer. Materials not included in the library were inserted according to the specifications provided by the customer.

2.1.1 Materials in the analyzed sections

Tab. 1 shows the characteristics of the materials used in the analysis.

Frame Materials	Conductivity (W/mK)	Emissivity	Source
6060 Aluminum alloy	160	0,9	Conductivity values from annex D, Tab. D1 of EN ISO 10077-2 Emissivity value from annex D, Tab. D3 of EN ISO 10077-2
6060 Aluminum alloy – slight oxidized aluminum surfaces	160	0,3	Conductivity values from annex D, Tab. D1 of EN ISO 10077-2 Emissivity value from annex D, Tab. D3 of EN ISO 10077-2
Aluminum Low-E Foil	160	0,02	Conductivity values from annex D, Tab. D1 of EN ISO 10077-2 Emissivity value provided by the applicant - Test Report No. 16-000059-PR01 (PB-H99-07-en-01) issued on 2016-02-19 by ift Rosenheim
EPDM	0,250	0,9	Conductivity values from annex D, Tab. D1 of EN ISO 10077-2 Emissivity value from EN ISO 10077-2
EPDM foam	0,050	0,9	Conductivity values from annex D, Tab. D1 of EN ISO 10077-2 Emissivity value from EN ISO 10077-2
Polyamide (Noryl)	0,180	0,9	Conductivity value provided by the applicant – Reports nr. 1754; 1773; 1775 of 04/011/20141 by Laboratory IIP S.r.l. Emissivity value from EN ISO 10077-2
PE Polyethylene foam	0,038	0,9	Conductivity value provided by the applicant - Test Report "Statement P1-266e/2015" issued on October 6, 2015, by Fraunhofer-Institut für Bauphysik IBP Emissivity value from EN ISO 10077-2

Tab. 1. Thermal characteristics of sample materials

3 Results

3.1 Calculation of thermal transmittance of frame nodes

IRCCOS TECHNICIAN	DATE OF CALCULATION	BOUNDARY CONDITIONS					
		Temperature		Internal surface resistance (R_{si})		External surface resistance (R_{se})	
		Inside (°C)	Outside (°C)	Normal (m^2K/W)	Reduced (m^2K/W)	Normal (m^2K/W)	Reduced (m^2K/W)
Rocco Musolino	2024-01-22	20	0	0,13	0,20	0,04	0,04

Tab. 2.

Fig. 2 shows the results obtained from the analyses performed on the nodes. The following are presented:

- the analyzed section with an indication of the materials, cavities and boundary conditions used;
- legend of materials used with indication of thermal conductivity and emissivity values;
- temperature trends within the analyzed section, with the corresponding legend;
- Heat flow trend within the analyzed section, with the corresponding legend;
- result obtained in W/m^2K rounded to the second significant figure.

3.1.1 Results on Node 1 - Lateral / Lower / Upper Node

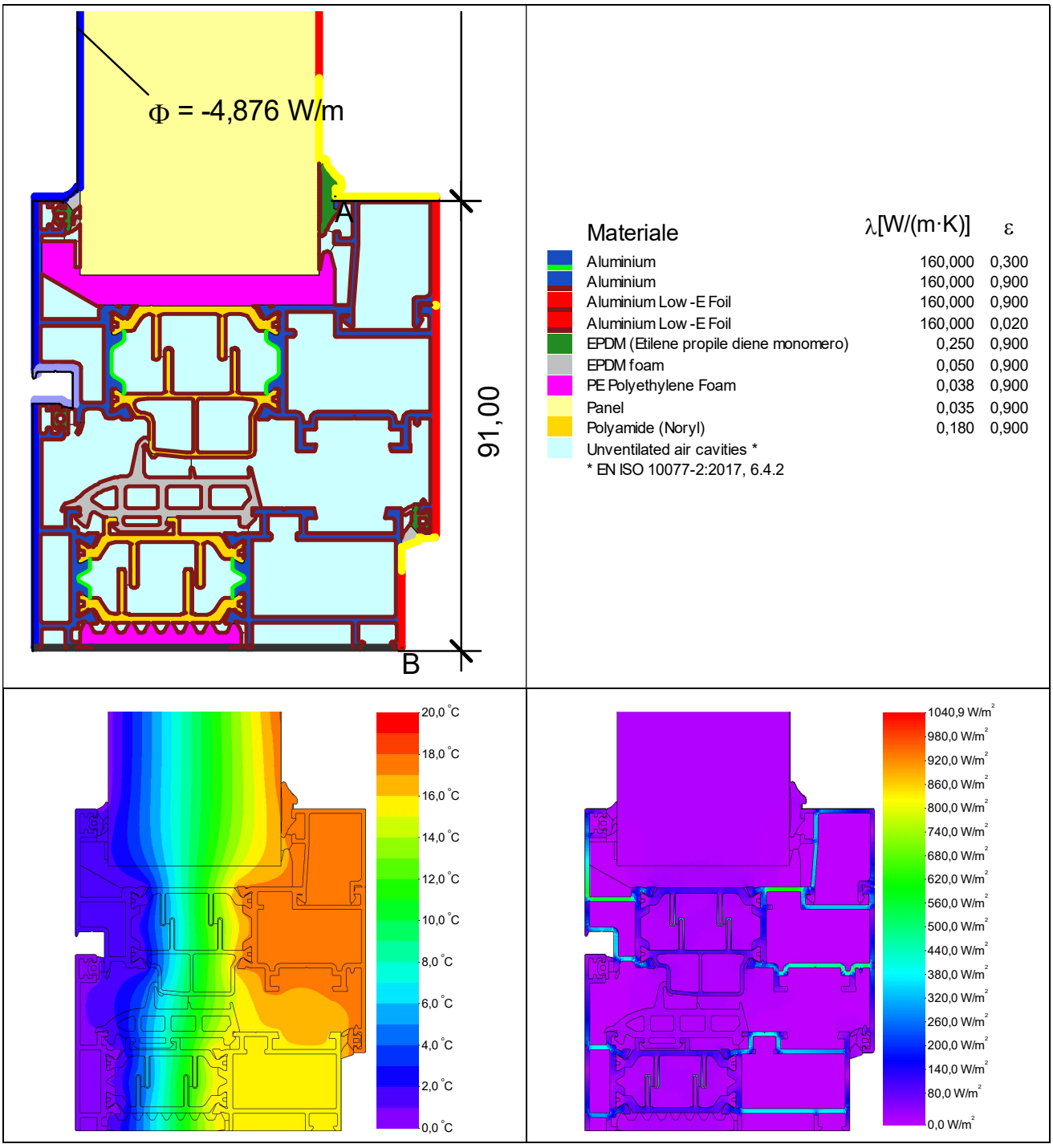


Fig. 2. Temperature and heat flow trends in the node 1 - Lateral / Lower / Upper Node

Thermal transmittance of node 1 - Lateral / Lower / Upper Node:

$U_f = 1{,}3\ W/m^2K$

4 Overview of results

SAMPLE	DESCRIPTION	RESULT VALUE (W/m²K)
Node 1	Lateral / Lower / Upper node	1,3

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Rocco Musolino

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